

FANUC Robot **series**
R-30iB/ R-30iB Plus CONTROLLER
Purged Process Enclosure and
Purged Motor Boxes
Operator's Manual

MAROBPPEB03201E REV A

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- **Original Instructions**

Thank you very much for purchasing FANUC Robot.
Before using the Robot, be sure to read the "FANUC Robot SAFETY HANDBOOK (B-80687EN)" and understand the content.

- No part of this manual may be reproduced in any form.
- The appearance and specifications of this product are subject to change without notice.

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In this manual, we endeavor to include all pertinent matters. There are, however, a very large number of operations that must not or cannot be performed, and if the manual contained them all, it would be enormous in volume. It is, therefore, requested to assume that any operations that are not explicitly described as being possible are "not possible".

SAFETY PRECAUTIONS

This chapter must be read before using the robot.

For detailed functions of the robot operation, read the relevant operator's manual to understand fully its specification.

For the safety of the operator and the system, follow all safety precautions when operating a robot and its peripheral equipment installed in a work cell.
For safe use of FANUC robots, you must read and follow the instructions in “FANUC Robot SAFETY HANDBOOK (B-80687EN)”.

1 DEFINITION OF USER

The personnel can be classified as follows.

- Operator:
- Turns the robot controller power on/off
 - Starts the robot program from operator panel

- Programmer or Teaching operator:
- Operates the robot
 - Teaches the robot inside the safety fence

- Maintenance technician:
- Operates the robot
 - Teaches the robot inside the safety fence
 - Performs maintenance (repair, adjustment, replacement)

- Operator is not allowed to work in the safety fence.
- Programmer/Teaching operator and maintenance technician is allowed to work in the safety fence. Works carried out in the safety fence include transportation, installation, teaching, adjustment, and maintenance.
- To work inside the safety fence, the person must be trained on proper robot operation.

SAFETY PRECAUTIONS

Table 1 (a) lists the work outside the safety fence. In this table, the symbol “○” means the work allowed to be carried out by the worker.

Table 1 (a) List of work outside the fence

	Operator	Programmer or Teaching operator	Maintenance technician
Turn power ON/OFF to Robot controller	○	○	○
Select operating mode (AUTO, T1, T2)		○	○
Select remote/local mode		○	○
Select robot program with teach pendant		○	○
Select robot program with external device		○	○
Start robot program with operator's panel	○	○	○
Start robot program with teach pendant		○	○
Reset alarm with operator's panel		○	○
Reset alarm with teach pendant		○	○
Set data on teach pendant		○	○
Teaching with teach pendant		○	○
Emergency stop with operator's panel	○	○	○
Emergency stop with teach pendant	○	○	○
Operator's panel maintenance			○
Teach pendant maintenance			○

In the robot operating, programming and maintenance, the operator, programmer/teaching operator and maintenance technician take care of their safety using at least the following safety protectors.

- Use clothes, uniform, overall adequate for the work
- Safety shoes
- Helmet

2 DEFINITION OF SAFETY NOTATIONS

To ensure the safety of users and prevent damage to the machine, this manual indicates each precaution on safety with "WARNING" or "CAUTION" according to its severity. Supplementary information is indicated by "NOTE". Read the contents of each "WARNING", "CAUTION" and "NOTE" before using the robot.

Symbol	Definitions
 WARNING	Used if hazard resulting in the death or serious injury of the user will be expected to occur if he or she fails to follow the approved procedure.
 CAUTION	Used if a hazard resulting in the minor or moderate injury of the user, or equipment damage may be expected to occur if he or she fails to follow the approved procedure.
NOTE	Used if a supplementary explanation not related to any of WARNING and CAUTION is to be indicated.

- Check this manual thoroughly, and keep it handy for the future reference.

1. PAINT ROBOT SAFETY

Today’s highly automated paint booths require that process and maintenance personnel have full awareness of the system and its capabilities. They must understand the interaction that occurs between any moving machinery along the conveyor and the robot(s), hood/deck and door opening devices, and high–voltage electrostatic tools.

Whenever personnel are working inside the paint booth, ventilation equipment must be used. Instruction on the proper use of ventilating equipment is usually provided by the paint shop supervisor.

 **CAUTION**
Ensure that all ground cables remain connected. Never operate the paint robot with ground provisions disconnected. Otherwise, you could injure personnel or damage equipment.

Paint robots are operated in three modes:

- Teach or manual mode
- Automatic mode, including automatic or Production mode
- Diagnostic mode

During both teach and automatic modes, the robots in the paint booth will follow a predetermined pattern of movements. In teach mode, the process technician teaches (programs) paint paths using the teach pendant.

In automatic mode, robot operation is operated at full production speed. It is initiated at the System Control Console (SCC) or Manual Control Panel (MCP), if available, and can be monitored from outside the paint booth. All personnel must remain outside of the booth whenever automatic mode is initiated at the SCC or MCP.

1.1 Paint System Safety Features

Process technicians and maintenance personnel must become familiar with the equipment and its capabilities. To minimize the risk of injury when working near robots and related equipment, personnel must comply strictly with the procedures in the manuals.

This section provides information about the safety features that are included in the paint system and also explains the way the robot interacts with other equipment in the system.

The paint system includes the following safety features (in addition to other standard robot safety features):

- Most paint booths have red warning beacons that illuminate when the robots are armed and ready to paint. Your booth might have other kinds of indicators. Learn what these are.
- Some paint booths have a blue beacon that, when illuminated, indicates that the electrostatic devices are enabled. Your booth might have other kinds of indicators. Learn what these are.
- An intrinsically safe teach pendant is used when teaching in hazardous paint atmospheres.
- Additional safety components for Paint systems such as Light curtains, door switches etc. are defined by a Risk Assessment.
- System safety devices are monitored by hardware circuits and/or a safety PLC. The robots are interlocked to the hardware circuits and/or safety PLC.

1.2 Staying Safe While Operating the Paint Robot

When you work in or near the paint booth, observe the following rules, in addition to all rules for safe operation that apply to all robot systems.

 **WARNING**
Never bypass, strap, or otherwise deactivate the robot purge system for any reason. Deactivating the purge system might result in an explosion serious injury or death.

 **WARNING**
Enclosures shall not be opened unless the area is known to be nonhazardous or all power has been removed from devices within the enclosure. Power shall not be restored after the enclosure has been opened until all combustible dusts have been removed from the interior of the enclosure and the enclosure purged. Refer to the Purge chapter for the required purge time.

- Know the work area of the entire paint station.
- Know the work envelope of the robot and hood/deck and door opening devices.
- Know the location and status of all switches, sensors, and/or control signals that might cause the robot, conveyor, and opening devices to move.
- Remove all metallic objects, such as rings, watches, and belts, before entering a booth when the electrostatic devices are enabled.
- Be aware of signals and/or operations that could start up paint process application equipment such as guns or bells.
- Be aware of potential air pressure buildup behind paint robot access covers.
- Be aware of all safety precautions when dispensing of paint is required.

1.3 Special Precautions for Combustible Dusts (Powder Paint)

When the robot is used in a location where combustible dusts are found, such as the application of powder paint, the following special precautions are required to insure that there are no combustible dusts inside the robot.

- Purge maintenance air should be maintained at all times, even when the robot power is off. This will insure that dust cannot enter the robot.
 - A purge cycle will not remove accumulated dusts. Therefore, if the robot is exposed to dust when maintenance air is not present, it will be necessary to remove the covers and clean out any accumulated dust.
 - Do not energize the robot until you have performed the following steps.
- (1) Before covers are removed, the exterior of the robot should be cleaned to remove accumulated dust.
 - (2) When cleaning and removing accumulated dust, either on the outside or inside of the robot, be sure to use methods appropriate for the type of dust that exists. Usually lint free rags dampened with water are acceptable. Do not use a vacuum cleaner to remove dust as it can generate static electricity and cause an explosion unless special precautions are taken.
 - (3) Thoroughly clean the interior of the robot with a lint free rag to remove any accumulated dust.
 - (4) When the dust has been removed, the covers must be replaced immediately.
 - (5) Immediately after the covers are replaced, run a complete purge cycle. The robot can now be energized.

1.4 Staying Safe While Operating Paint Application Equipment

When you work with paint application equipment, observe the following rules, in addition to all rules for safe operation that apply to all robot systems.

 **WARNING**
When working with electrostatic paint equipment, follow all national and local codes as well as all safety guidelines within your organization. Also reference the following standards: NFPA 33 Standards for Spray Application Using Flammable or Combustible Materials, and NFPA 70 National Electrical Code.

Grounding: All electrically conductive objects in the spray area must be grounded. This includes the spray booth, robots, conveyors, workstations, part carriers, hooks, paint pressure pots, as well as solvent containers. Grounding is defined as the object or objects shall be electrically connected to ground with a resistance of not more than 1 megohms.

High Voltage: High voltage should only be on during actual spray operations. Voltage should be off when the painting process is completed. Never leave high voltage on during a cap cleaning process.

- (1) Avoid any accumulation of combustible vapors or coating matter.
- (2) Follow all manufacturer recommended cleaning procedures.
- (3) Make sure all interlocks are operational.
- (4) No smoking.
- (5) Post all warning signs regarding the electrostatic equipment and operation of electrostatic equipment according to NFPA 33 Standard for Spray Application Using Flammable or Combustible Material.
- (6) Disable all air and paint pressure to the bell.
- (7) Verify that the lines are not under pressure.
- (8) Be aware of potential hazards presented by paint process hardware, such as rotating or spinning components on the paint applicator.

1.5 Staying Safe During Maintenance

When you perform maintenance on the painter system, observe the following rules, and all other maintenance safety rules that apply to all robot installations. Only qualified, trained service or maintenance personnel should perform repair work on a robot.

- (1) Paint robots operate in a potentially explosive environment. Use caution when working with electric tools.
- (2) Be sure all covers and inspection plates are in good repair and in place.
- (3) Whenever possible, turn off the main electrical disconnect before you clean the robot.
- (4) When using solvents, observe the following:
 - Wear eye protection and protective gloves during application and removal.
 - Adequate ventilation is required. Overexposure could cause drowsiness or skin and eye irritation.
 - If there is contact with the skin, wash with water.
 - Follow the Original Equipment Manufacturer’s Material Safety Data Sheets.

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1 OVERVIEW

1.1 Configurations

This manual supports the following Purged Motor Boxes (PMB) and Purged Process Enclosure (PPE) configurations:

PMB
Beta iS 0.5/6000 W/ BRAKE 10:1 reducer PMB
Beta iS 1.0/6000 7:1 ST SHAFT reducer PMB
Beta iS 1.0/6000 W/ BRAKE 10:1 ST SHAFT reducer PMB
Beta iS 1.0/6000 W/ BRAKE 10:1 SPL SHAFT reducer PMB
Alpha iS 4/5000 W/ BRAKE TAPERED SHAFT PMB
Alpha iS 8/4000 W/ BRAKE TAPERED SHAFT PMB
Alpha iS 12/4000 W/ BRAKE TAPERED SHAFT PMB
Alpha iS 12/4000 W/ BRAKE STRAIGHT SHAFT PMB
Alpha iS 30/3000 W/ BRAKE TAPERED SHAFT PMB
Alpha iS 40/4000 W/ BRAKE STRAIGHT SHAFT PMB
PPE
Blank Purged Process Enclosure Assembly
Basic Purged Process Enclosure Assembly
1K Bell Process Installation (PPE)
1K Gun Process Installation (PPE)
2K Bell Process Installation (PPE)
2K Gun Process Installation (PPE)

1.2 Purged Process Enclosure Restrictions

1.2.1 Purged Process Enclosure open architecture use restrictions

Purged Process Enclosures offer the ability to operate Process Hardware inside a Hazardous Location. The process hardware includes solenoids for applicator control, solenoids for color change control, transducers for atomizing air and bell speed, and analog inputs for pressure monitoring. There are standard Process Hardware configurations to support solventborne applications with the Versabell applicator. Variations of the SMC EX600 or other equipment are supported with the following limitations:

- All power supplied to the PPE must be redundantly purge supervised
- Not more than 4 IS circuits are to leave the purged cavity per the agency control drawing
- Power supplied to the PPE is not to exceed 100 watts
- Purged cavity penetrations must match the agency controlled drawings
- Emitted light must not exceed 10uW per fiber exiting into the hazardous location
- No additional holes in the enclosure are allowed
- Fittings for electrical and pneumatic penetrations must match the agency controlled drawing

1.3 Purge Air Supply Requirements

Table 1-1 Purge Air Supply Requirements

Maximum Internal Volume	199 Liters
Minimum Flow Switch Activation	566 SLPM
Maximum Flow Switch Activation	740 SLPM
Air Supply Pressure	448 - 517 kPa (65 - 75 psi) Dynamic
Maximum Air Supply Temperature	45°C
Maximum Maintenance Flow	70 - 410 SLPM
Maintenance Pressure	2.491 - 3.238 kPa (10 - 13 Inches of H ₂ O)
Minimum Overpressure	0.172 kPa (0.69 Inches of H ₂ O)
Maximum Overpressure	6.9 kPa (1.0 psi)
Maximum Leakage Flow	50 SLPM
Operating Ambient Temperature	0°C - 45°C
Humidity	-40°C Pressure Dewpoint at 621 kPa
Filtration	5 Micron Absolute
Exhaust Pressure	0.149 kPa (0.6 Inches of H ₂ O)

2 PPE AND PMB INSTALLATION

Figure 2-1 ET-6854-800-001, PPE AND PMB INSTALLATION

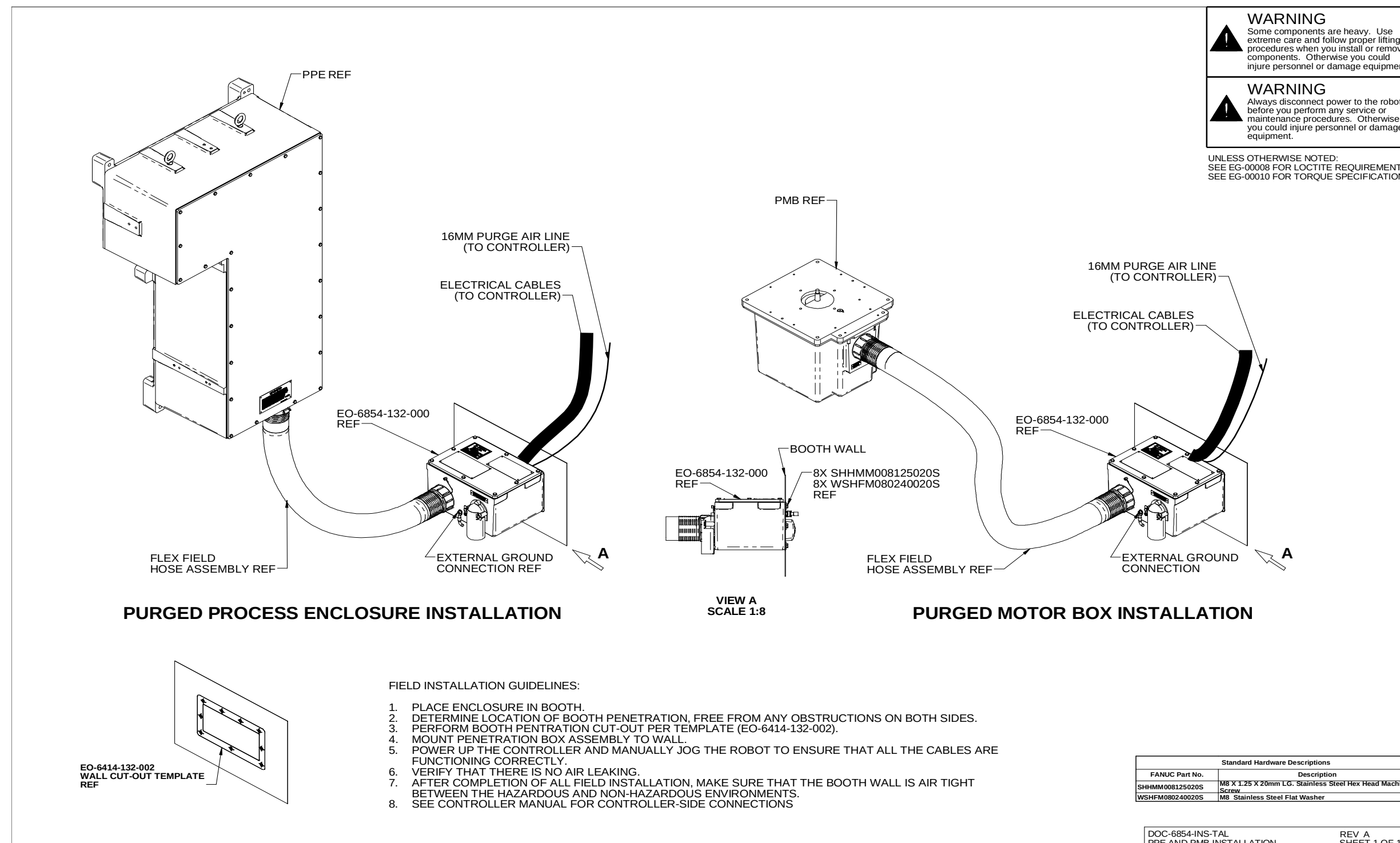
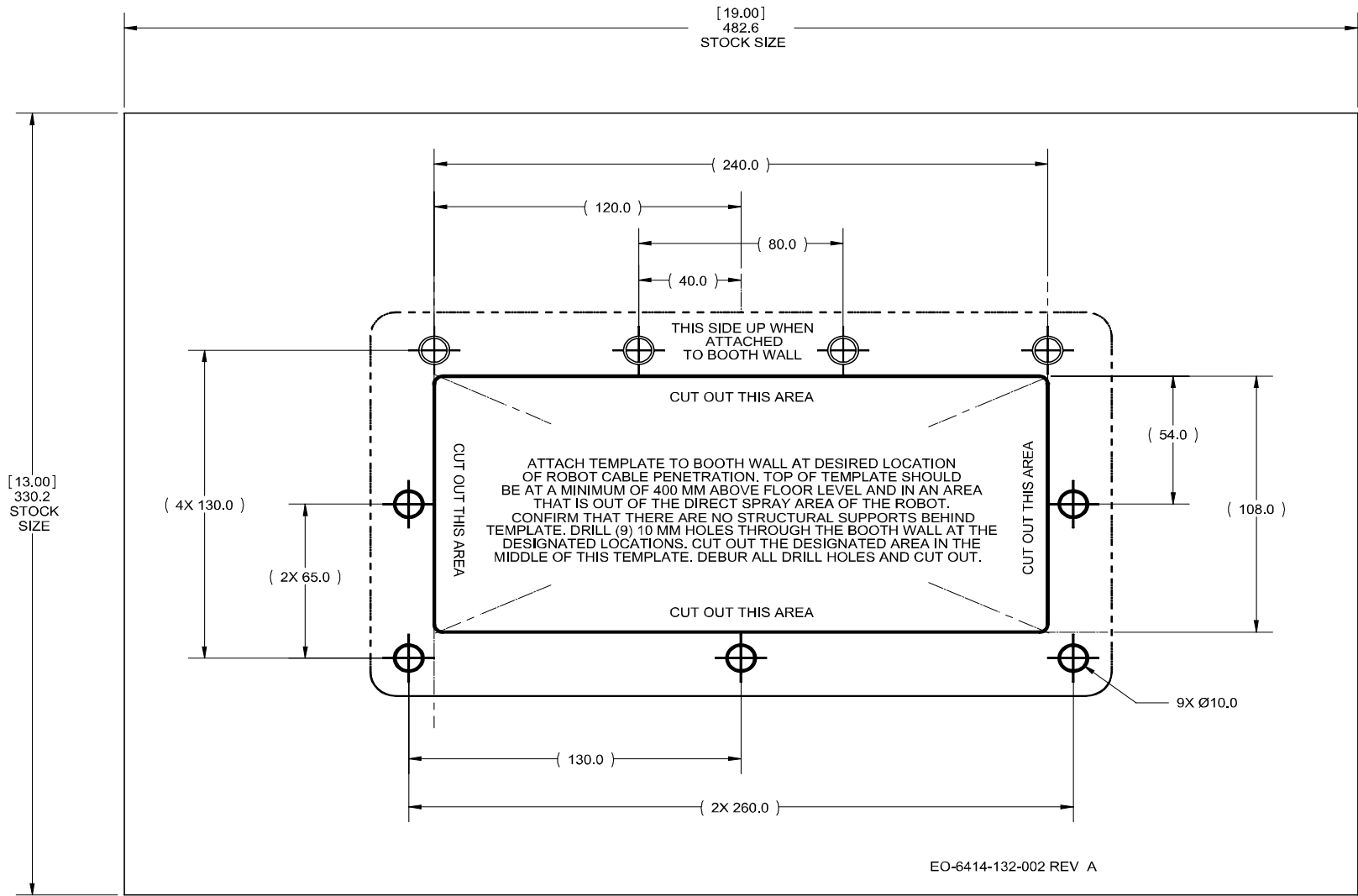


Figure 2-2 EO-6414-132-002, TEMPLATE BOOTH - WALL



WARNING
Some components are heavy. Use extreme care and follow proper lifting procedures when you install or remove components. Otherwise you could injure personnel or damage equipment.

WARNING
Always disconnect power to the robot before you perform any service or maintenance procedures. Otherwise, you could injure personnel or damage equipment.

UNLESS OTHERWISE NOTED:
SEE EG-00008 FOR LOCTITE REQUIREMENTS
SEE EG-00010 FOR TORQUE SPECIFICATIONS

REFERENCE ONLY. NOT TO SCALE.

Figure 2-3 ET-6854-800-002, PMB INTERFACE

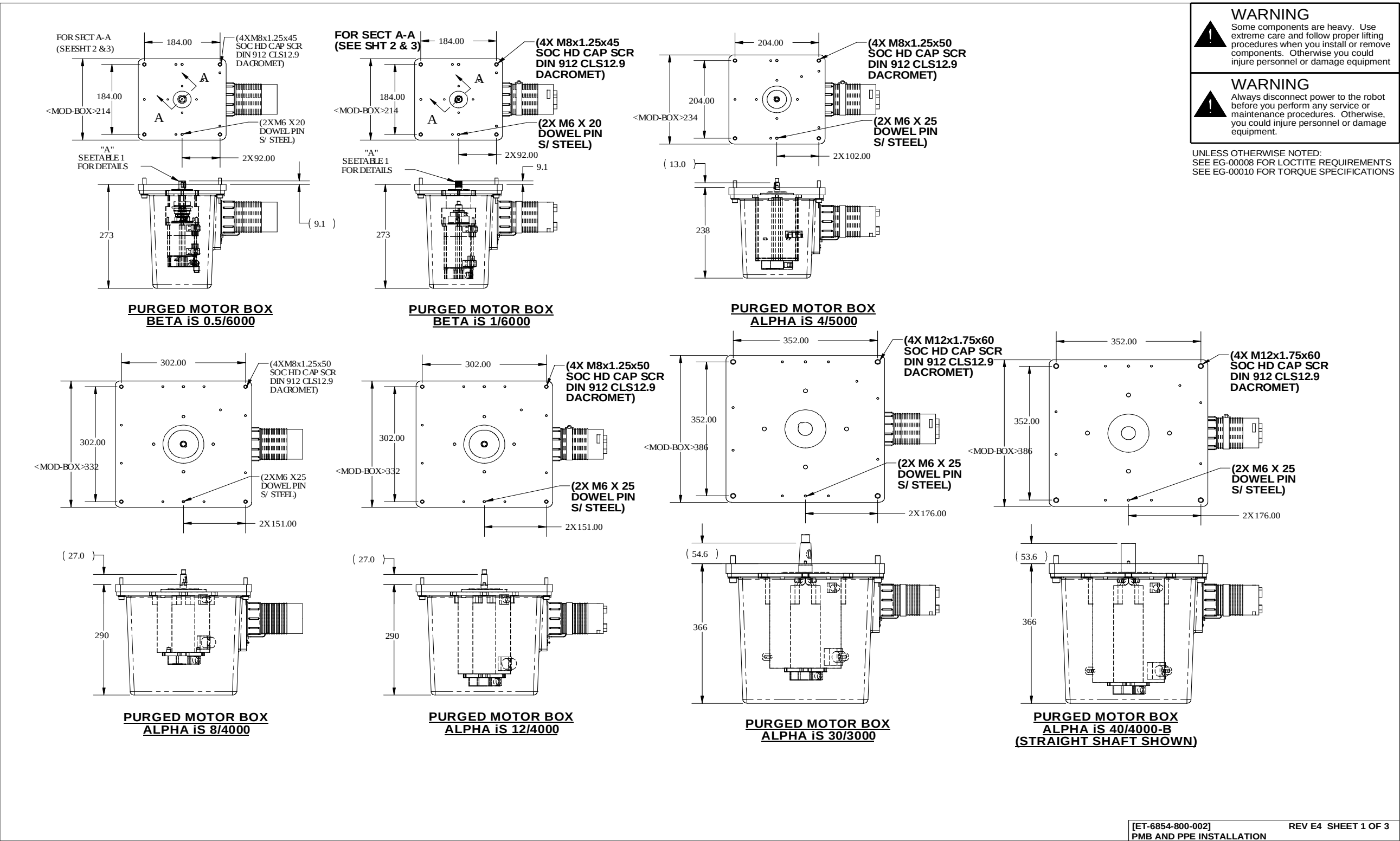


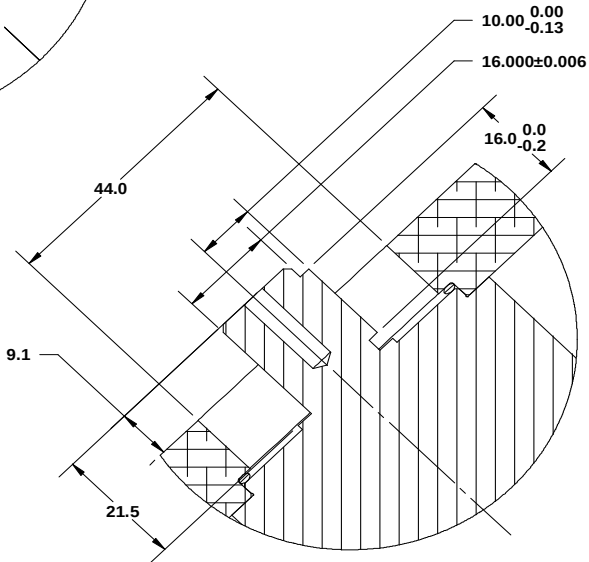
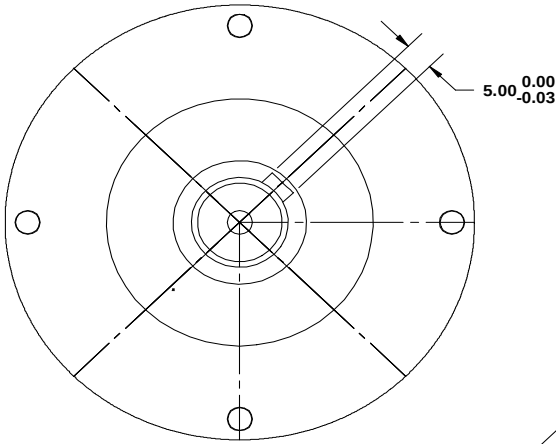
Figure 2-4 ET-6854-800-002, PMB INTERFACE

TABLE 1

PURGED MOTOR BOX ASM PART #.	MOTOR AND REDUCER DESCRIPTION	REDUCER RATIO	REDUCER SHAFT
EO6854-010-000A	BETA 0.5 W/ BRK	10:1	WITH KEY
EO6854-010-000B	BETA 1 W/ BRK	10:1	WITH KEY
EO6854-010-000C	BETA 1 W/ BRK	10:1	WITH SPLINE
EO6854-010-000D	BETA 1 W/ BRK	7:1	WITH KEY

TABLE 2

REDUCER PERFORMANCE SPECIFICATION		
	REDUCER RATIO 7:1	REDUCER RATIO 10:1
MAX ACCELERATION TORQUE [NM] (MAX. 1000 CYCLES PER HOUR)	32	29
NOMINAL OUTPUT TORQUE [NM]	16	15
EMERGENCY STOP TORQUE (PERMITTED 1000 TIMES DURING SERVICE LIFE OF THE GEARHEAD) [NM]	75	75
MAX. INPUT SPEED [RPM]	6000	6000
MEAN NO LOAD RUNNING TORQUE [NM] (WITH N=3000RPM AND 20°C GEARHEAD TEMPERATURE)	0.11	0.11
MAX. TORSIONAL BACKLASH [ARCMIN]	≤20	≤20
TORSIONAL RIGIDITY [NM/ARCMIN]	2.1	1.9
MAX. AXIAL FORCE [N]	750	750
MAX. RADIAL FORCE [N]	650	650
EFFICIENCY AT FULL LOAD [%]	97	97
MOMENT OF INERTIA (RELATES TO THE DRIVE) [KGC ^{M2}]	0.17	0.17



SECTION A-A
FROM SHEET 1
FOR PMB WITH SHAFT AND KEY, SEE TABLE 1 OPTION

WARNING

Some components are heavy. Use extreme care and follow proper lifting procedures when you install or remove components. Otherwise you could injure personnel or damage equipment

WARNING

Always disconnect power to the robot before you perform any service or maintenance procedures. Otherwise, you could injure personnel or damage equipment.

UNLESS OTHERWISE NOTED:
SEE EG-00008 FOR LOCTITE REQUIREMENTS
SEE EG-00010 FOR TORQUE SPECIFICATIONS

Figure 2-5 ET-6854-800-002, PMB INTERFACE

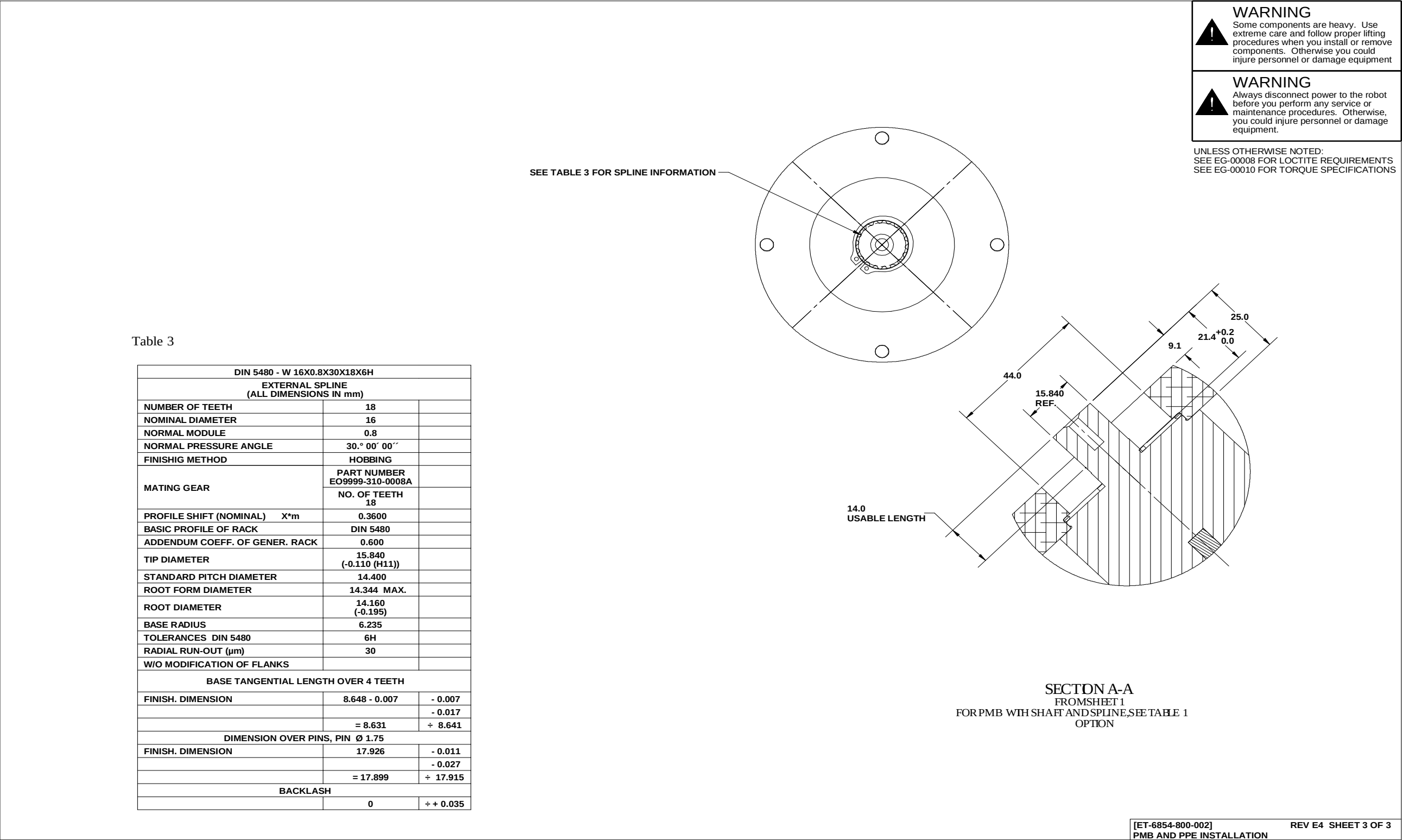
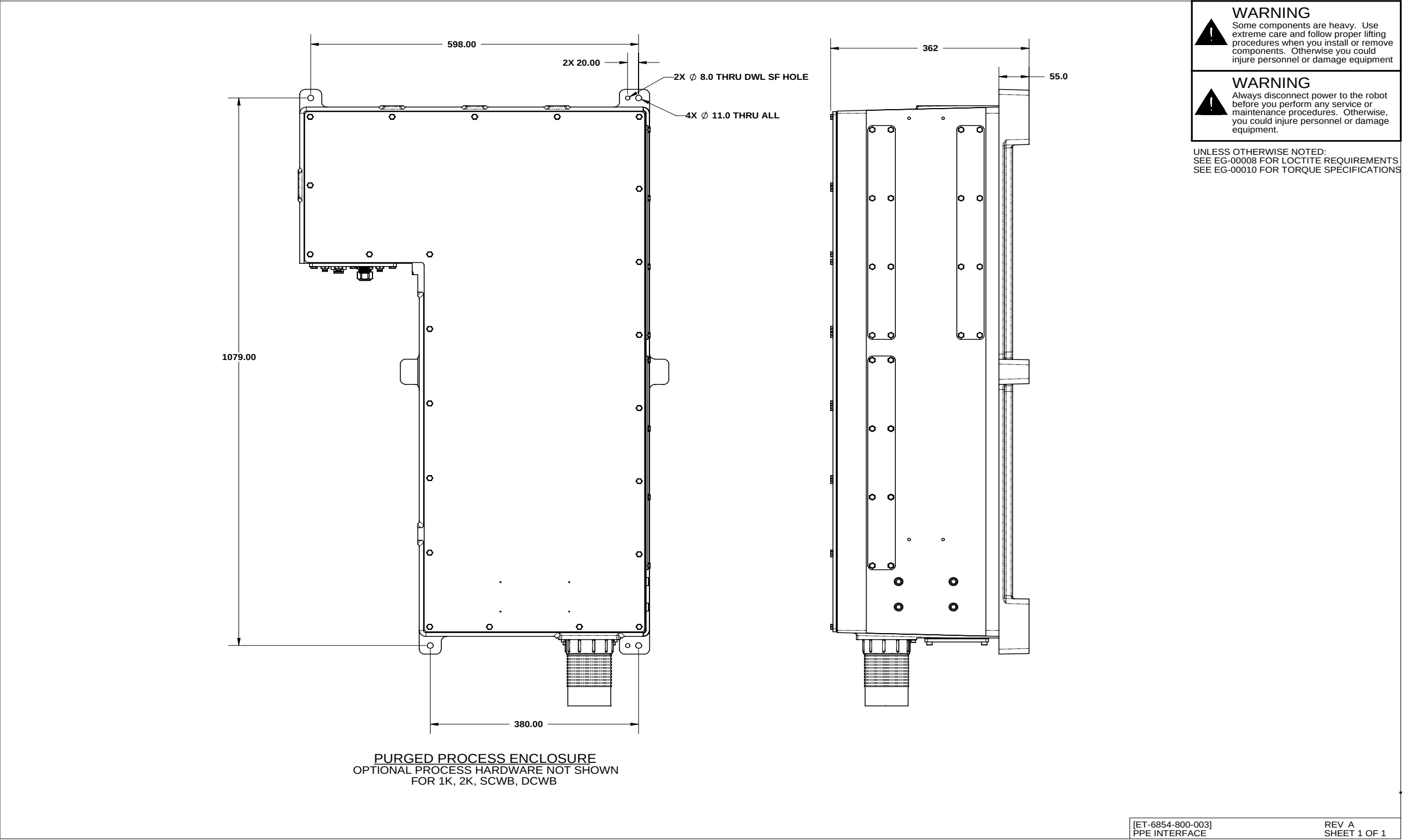


Figure 2-6 ET-6854-800-003, PPE INTERFACE



3 TROUBLESHOOTING

3.1 Mechanical Unit Troubleshooting

Table 3-1 shows the most common robot mechanical unit problems and their causes. If a cause or remedy is unclear, contact FANUC America Corporation.

Table 3-1 Problems and Causes

Symptom	Cause	Remedy
The robot does not return to the zero position correctly	Low voltage of the memory backup battery.	Replace the battery.
	Unusual external start signal.	Investigate the cause.
	Incorrect parameter.	Revise parameters.
Position offset	The robot has been hit.	Revise the teaching points.
	Incorrect Mastering	Remaster Robot
	The robot is not firmly fastened to the base plate.	Retighten.
	Peripheral device was shifted.	Reposition.
	Incorrect parameters.	Revise parameters.
	Defective pulse coder cable.	Replace the cable.
	APC (Absolute Pulse Coder) malfunction.	Replace the motor or the pulse coder.
	Play in mechanical unit.	Refer to excessive play symptom in this table.
Vibration	The robot is not firmly fastened to the base.	Retighten.
	Excessive floor vibration (especially when installed above the first floor).	Investigate the installation location and adjust the robot location if necessary.
	An incorrect servo adjustment.	Adjust the servo.
	A defective cable.	Replace the cable.
	The robot is not grounded.	Connect to ground.
	A defective motor.	Replace the motor.
	A defective reducer.	Replace the reducer.
	Play in the mechanical unit.	Refer to the excessive play symptom in this table.
Excessive play	A loose screw or pin.	Tighten and apply LOCTITE as specified.
	A defective reducer.	Replace the reducer.
	An incorrect gear backlash.	Adjust the gears.
	Worn gears.	Replace the gears.
	Worn bearings.	Replace the bearings.
Unusual noise	Insufficient gear or reducer lubrications.	Lubricate.
	Foreign matter in a gear or a reducer.	Clean and lubricate.
	Excessive pressure on the bearings.	Reduce the pressure on the bearings.
	A defective reducer.	Replace the reducer.
	An incorrect gear adjustment.	Adjust the gears.
	Worn gears.	Replace the gears.
	Worn bearings.	Replace the bearings.
	Cable pair damage.	Replace the cable pair.
	An incorrect servomotor adjustment.	Adjust the servomotor.
Over heating	Insufficient gear or reducer lubrication.	Lubricate.
	The specified grease was not used.	Replace with the specified grease.
	Excessive pressure on bearings.	Reduce the pressure on the bearings.
	Insufficient maintenance air	Increase maintenance air supply
	An excessive load on the wrist.	Reduce the load and restrict the movement conditions.
	An incorrect gear adjustment.	Adjust the gears.
	An incorrect time constant.	Revise the time constant.

Table 3-1 Problems and Causes, continued

Symptom	Cause	Remedy
The robot drops (10 mm or more) when the power supply is off	The brake gap is too large.	Replace the motor.
	Fused brake drive relay.	Replace the relay.
Grease leakage	Deteriorated O-ring, oil seal, or packing.	Replace the O-ring, oil seal, or packing.
	A cracked casting.	Replace the cracked parts.
	Loose screws.	Tighten the screws.
	Excessive quantity of grease.	Remove some of the grease.

4 ENGINEERING GUIDELINES

4.1 Loctite Spec - EG-00008

Figure 4-1 EG-00008 Sheet 1 of 10, LOCTITE APPLICATION

FANUC AMERICA CORPORATION **ENGINEERING GUIDELINE**

Page 1 of 10

Title: LOCTITE APPLICATION

EG #: EG-00008

The purpose of this engineering guideline is to define the different Loctite products used at FAC and to provide directions for proper use.

This document is intended as a guideline only. Deviations are permitted and override this guideline when explicitly stated on engineering drawings/documentation.

1. Cleaning

Surfaces that are to have Loctite applied must be clean.

Acetone and Loctite "ODC-Free Cleaner & Degreaser" are approved cleaning agents.

Fasteners may be batch cleaned by submerging in acetone.

Critical areas where retaining compounds are used must be thoroughly cleaned to remove all preservative. Some preservatives are not visible and the parts may appear clean. Some preservatives contain nitrates which will stop Loctite from curing. All areas where retaining compound is used will have it's final cleaning with acetone.

The parts on which retaining compounds are used shall be thoroughly cleaned and "scrubbed" with acetone at least two times with a clean cloth. They may be washed beforehand in a solvent tank. The parts must then be wiped down and blown off before cleaning twice with acetone.

2. Primers

The engineering drawings shall call out primer when necessary.

Loctite 7649 (Primer N) ND 611 & Loctite 7471 (Primer T) ND 612

Loctite brand primers are used to ensure cure when applying Loctite threadlocker to two inactive metals. Primers will also speed up cure time.

Figure 4-2 EG-00008 Sheet 2 of 10, LOCTITE APPLICATION

FANUC AMERICA CORPORATION **ENGINEERING GUIDELINE**

Title: LOCTITE APPLICATION

Page 2 of 10

EG #: EG-00008

Active Metals	Inactive Metals
Iron	Plated Steel
Plain Steel	Anodized Aluminum
Copper	Stainless Steel
Brass	Galvanized Steel
Commercial Aluminum	Zinc
Bronze	Natural or Chemical Black Oxide
Kovar	Plastic
	Dacromet

If a primer is required for faster curing, it should be specified on the drawing.

Directions for use

- 1) Clean mating surfaces to ensure all contamination has been removed.
- 2) Apply primer to both mating surfaces to be bonded.
- 3) Allow the primer time to evaporate under good ventilation until surfaces are completely dry.
- 4) After primer is applied, parts should be bonded within 7 days.

3A. Thread Locking – Conventional.

Loctite thread lockers are anaerobic and cure when confined in the absence of air between close fitting metal surfaces. Application of the product will prevent loosening of assembled parts that are subject to shock and vibration.

Loctite 222 ND 111:

Figure 4-3 EG-00008 Sheet 3 of 10, LOCTITE APPLICATION

FANUC AMERICA CORPORATION

ENGINEERING GUIDELINE

Title: LOCTITE APPLICATION

Page 3 of 10
EG #: EG-00008

Loctite 222 is a low strength threadlocker designed to lock and seal threaded fasteners, which require easy disassembly. Unless explicitly stated on engineering drawings/documentation, Loctite 222 is to be used on all carbon and stainless steel fasteners 5mm and smaller. The uncured appearance of this product is a purple liquid.

Loctite 242 ND 121 or 248(stick):

Loctite 242 is a medium strength treadlocker designed to lock and seal threaded fasteners that require normal disassembly with standard hand tools. Unless explicitly stated on engineering drawings/documentation, Loctite 242 is to be used on all carbon and stainless steel fasteners 6mm and above. The uncured appearance of this product is blue.

Loctite 262 ND 131 or 268(stick):

Loctite 262 is a high strength threadlocker designed to permanently lock and seal threaded fasteners. This product should not be applied unless explicitly stated on engineering drawings/documentation. The uncured appearance of this product is red.

Loctite 277 ND 146:

Loctite 277 is a high strength threadlocker designed to permanently lock and seal threaded fasteners. This product should not be applied unless explicitly stated on engineering drawings/documentation. The uncured appearance of this product is a red liquid.

Directions for use

- 1) Clean mating surfaces to ensure all contamination has been removed.
- 2) Apply primer as required and allow to dry.
- 3) Select the specified product. Shake the liquid products before use.
- 4) **For Thru Holes**, apply several drops of the product onto the bolt at the nut engagement area.
- 5) **For Blind Holes**, the application method requires that product be applied to both parts. Apply several drops of the product down the internal threads to the bottom of the hole as well as the external treads of mating fastener. If product

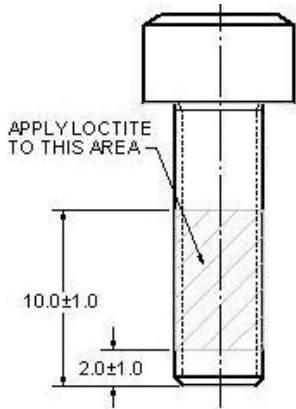
Figure 4-4 EG-00008 Sheet 4 of 10, LOCTITE APPLICATION

is only applied to the fastener, air pressure will force the liquid threadlocker to escape as the fastener is tightened down. Lack of uniform coverage creates air pockets, causing incomplete cures, which lead to failures.

6) Assemble and tighten as required. Fasteners must be torqued within 3 minutes.

3B. Thread Locking – Pre applied.

Loctite Dri Loc STS can be pre applied to fasteners provided the break away torque meets the FAC torque specification EG -00010.
For M5 fasteners it should be diluted 50 % by weight and applied as shown below.



4. Retaining

Loctite retaining compounds form a strong bond between press-fitted and slip-fitted parts. The product fills the inner space between components and cures to form a strong bond.

Loctite 680 is a high viscosity-retaining compound for joining fitted cylindrical parts. The product cures when confined in the absence of air between close fitting metals.

Figure 4-5 EG-00008 Sheet 5 of 10, LOCTITE APPLICATION

FANUC AMERICA CORPORATION

ENGINEERING GUIDELINE

Title: LOCTITE APPLICATION

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EG #: EG-00008

Directions for use

- 1) Clean mating surfaces to ensure all contamination has been removed. Parts must be cleaned completely. Use acetone and scrub with a clean rag at least two times.
- 2) If parts are inactive metals or cure speed is too slow, spray with Loctite Primer 7471 or 7649 per the engineering drawing.
- 3) **For Slip Fitted Assemblies**, apply adhesive 360 degrees around OD and ID of mating parts. Use a rotating motion during assembly to ensure good coverage.
- 4) **For Press Fitted Assemblies**, apply adhesive 360 degrees around OD and ID of mating parts to insure the surfaces are completely coated during assembly. Assemble mating parts. When an undercut is provided, the adhesive shall be applied to the shaft to insure that the undercut is completely filled as the parts are pressed together.
- 5) Parts should not be disturbed until sufficient handling strength is achieved. Parts using Loctite 680 need to remain in fixture for 10 minutes. Full cure will be developed in 24 hours.

5a. Thread Sealant

Loctite “No More Leaks” is a thread sealant, which is designed to be used on tapered threads of plastic fittings that carry water or air. The sealant is solvent based and is non-curing so it shouldn’t be used for lines that carry paint or solvents. It is not a thread locker

Directions for use

- 1) Clean mating surfaces to ensure contamination has been removed.
- 2) Apply a 360-degree bead of sealant to the leading threads of the male fitting, leaving the first thread free. Force the sealant into the threads to ensure all voids are filled.
- 3) Assemble and tighten fittings as required.
- 4) Properly tightened fittings will seal instantly because it does not actually cure.

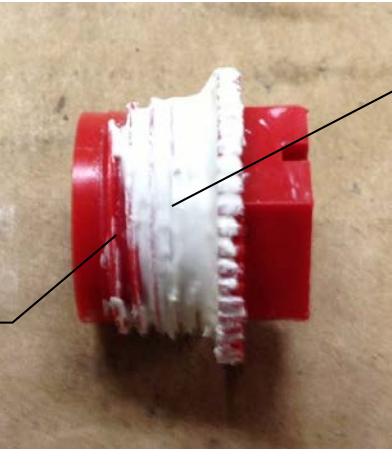
Figure 4-6 EG-00008 Sheet 6 of 10, LOCTITE APPLICATION

FANUC AMERICA CORPORATION
Title: LOCTITE APPLICATION

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First thread
free of sealant



100% coverage, sealant
forced into thread to
eliminate voids.

Fitting install,
sealant should be
visible at the edges.



Figure 4-7 EG-00008 Sheet 7 of 10, LOCTITE APPLICATION

5b. Thread Sealant and locker

A thread sealant will seal and secure metal and plastic pipes and fittings, filling the space between threaded metal parts, and hardening to prevent leakage. It can be used with primer on any metal and plastic fittings that are resistant to solvents such as Delrin (Acetal) or Nylon. The primer contains Acetone so it shouldn't be used on any plastic such as PVC that isn't resistant to paint solvents.

Loctite 567 (ND 460) is designed for locking and sealing metal tapered threads and fittings. The product cures when confined in the absence of air between close fitting metal surfaces and prevents loosening and leakage from shock and vibration. The product also provides lubrication, preventing galling between metal pipe threads and fittings.

Loctite 554 (ND 446) is similar Loctite 567 (ND 460). It should be used for locking and sealing of threads where maximum solvent resistance is required such as hoses and pipes that carry paint and paint solvents.

Directions for use

- 1) Clean mating surfaces to ensure contamination has been removed.
- 2) Apply primer to both the male and female threads as required and allow them to dry.
- 3) Apply a 360-degree bead of sealant to the leading threads of the male fitting, leaving the first thread free. Force the sealant into the threads to ensure all voids are filled.
- 4) Assemble and tighten fittings as required.
- 5) Properly tightened fittings will seal instantly and can withstand moderate pressure. For maximum pressure and solvent resistance allow the sealant to cure a minimum of 24 hours.

6. Gasketing

Gaskets are used to prevent fluid or gas leaks by forming impervious barriers that must remain intact and leak free for a prolonged period of time. Form-in-place gaskets are applied as a liquid sealant before parts are assembled. After

Figure 4-8 EG-00008 Sheet 8 of 10, LOCTITE APPLICATION

assembly, the form-in-place gasket spreads and cures between assembled parts filling gaps, scratches and surface irregularities to provide a durable seal.

Loctite 518 ND 730 Gasket Eliminator is medium strength, solvent resistant flexible seal that will not tear or decay. The product cures when confined in the absence of air between close fitting metals. The product will fill gaps up to 0.25mm.

Directions for use

- 1) Clean mating surfaces to ensure all contamination has been removed. The product gives best performance when surfaces are clean and free of grease.
- 2) Apply product in a continuous bead to one of the mating surfaces.
- 3) Assemble mating parts. Parts should be tightened as soon as possible after assembly.
- 4) Low pressure can be used to confirm a complete seal immediately after assembly. Allow time for full cure before applying high pressure.

7. Gasket Adhesives

Gasket adhesives are used to bond a gasket to its cover for ease of assembly.

Loctite "High Tack" is a flexible material that will bond gaskets to its cover. It can be used on all paper, rubber, cork or metal gaskets.

Directions for use

- 1) Clean mating surfaces to ensure all contamination has been removed.
- 2) Apply product with brush or spray.
- 3) Assemble gasket to cover. Parts should be allowed to cure for 5 minutes minimum before the cover/gasket is installed.

Figure 4-9 EG-00008 Sheet 9 of 10, LOCTITE APPLICATION

FANUC AMERICA CORPORATION ENGINEERING GUIDELINE

Title: LOCTITE APPLICATION

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- 4) After each cover removal, verify proper gasket attachment to the cover and reapply Permatex Hy-Tack HDWMO0000036300 if necessary.

8. List of the ND Industries part numbers – equivalent to Loctite.

Fanuc Item Number:	Current Item Description	New Item Description
HDWMO000001145O	LOCTITE #49550 ADHSV 1 OZBOTL	LOCTITE #49550 1oz / ND #39528
HDWMO000001146O	LOCTITE #242-41 SEALANT 250CC	LOCTITE #242-41 250cc/ND#12125
HDWMO000001147O ALT:HDWMO000002362O	LOCTITE #271-41 THRDLOCKR 250CC	LOCTITE #271-41 250cc/ND#14025
HDWMO000001836O	LOCTITE #242-31 SEALANT 50CC	LOCTITE #242-31 50cc/ND #12150
HDWMO000002312O	LOCTITE #515-31 GSKT SEALANT	LOCTITE #515-31 50cc/ND #73250
HDWMO000002362O ALT:HDWMO000001147O	LOCTITE #271-31 SEALANT 50CC	LOCTITE #271-31 50cc/ND #14050
HDWMO000002450O	LOCTITE #26231 RETAIN COMPOUND	LOCTITE #26231 50cc/ND #13150
HDWMO000003076O	LOCTITE #680 RETAIN COMP 50CC	LOCTITE #680 50cc/ND #54150
HDWMO000003148O	LOCTITE #290-41 SEALANT 250CC	LOCTITE #290-31 50cc/ND#15050
HDWMO000003185O	LOCTITE #PST- 56747 SEALANT 50M	LOCTITE#PST56747 50ml/ND#46050
HDWMO000003186O	LOCTITE #22241 SEALANT 250CC	LOCTITE #22241 250cc/ND #11125

Figure 4-10 EG-00008 Sheet 10 of 10, LOCTITE APPLICATION

FANUC AMERICA CORPORATION ENGINEERING GUIDELINE Title: LOCTITE APPLICATION Page 10 of 10 EG #: EG-00008		
HDWMO000003248O	LOCTITE #51831 SEALANT TUBE	LOCTITE #51831 50ml/ND #73050
HDWMO000003249O	LOCTITE #51845 SEALANT 300ML	LOCTITE #51845 300ml/ND#73025
HDWMO000003367O	LOCTITE #222-31 SEALANT 50 ML	LOCTITE #222-31 50ml/ND #11150
HDWMO000003658O	LOCTITE #21348 PRIMER N	LOCTITE #21348 PRI N/ND #61118
HDWMO0000060324O	LOCTITE #45440 PRISM 454 ADHES	LOCTITE #45440 20gm/ND #35420
HDWMO0000056588O	LOCTITE #41650 416 INST ADHEV	LOCTITE #41650 1oz/ND #31628
HDWMO0000057262O	Loctite #55441 Thread Sealant Locker	Loctite #55441 250cc ND #44625
HDWMO0000057263O	ND #44650 50cc Same as Loctite #554	ND #44650 50cc

4.2 Torque Spec - EG-00010

Figure 4-11 EG-00010 Sheet 1 of 5, TORQUE SPECIFICATION

FANUC AMERICA CORPORATION **ENGINEERING GUIDELINE**

Page 1 of 5

Title: Mechanical Development Torque Specifications

EG #: EG-00010

Purpose: To define the torque and Loctite specifications for the most common fasteners used.

Scope: This torque specification applies only to Products designed in the Mechanical Development department (EO, ET drawing series). Deviations are permitted and override this specification when explicitly stated on the engineering drawing/documentation.

Tolerance: - Torque tolerance for all specified values is assumed to be +/- 6% unless otherwise specified on drawing.

Specification: For Metric Fasteners Only

NOTE: Unless otherwise specified:

All threaded fasteners have coarse threads

All Stainless Socket Head fasteners and Stainless Hex Head screws and all Hex Head Type Nuts will be grade A2.

All Carbon Steel Socket Cap screws will be grade 12.9.

All Dacromet plated Carbon Steel Socket Cap screws will be grade 12.9.

All zinc plated Carbon Steel Socket Cap screws will be grade 12.9. (torqued to 10.9 spec)

All Carbon Steel Shoulder Screws will be grade 12.9.

All Carbon Steel Socket Flat Head screws and Carbon Steel Socket Button Head fasteners will be grade 10.9.

All Carbon Steel Low Head Socket Cap screws will be grade 8.8.

All Carbon Steel Hex head screws will be grade 8.8

All Carbon Steel Hex nuts M6 and larger will be grade 10

All Carbon Steel Hex Nut M5 and smaller will be grade 6

All Carbon Steel Jam Nuts will be grade 4.

All Nylon fasteners are Polyamide 6.6 (nylon 6.6)

All Brass fasteners are Brass Ms 63 (cold formed)

Loctite #222 will be used on all carbon and stainless steel fasteners 5mm and under.

Loctite #242 will be used on all carbon and stainless steel fasteners 6mm and over.

Figure 4-12 EG-00010 Sheet 2 of 5, TORQUE SPECIFICATION

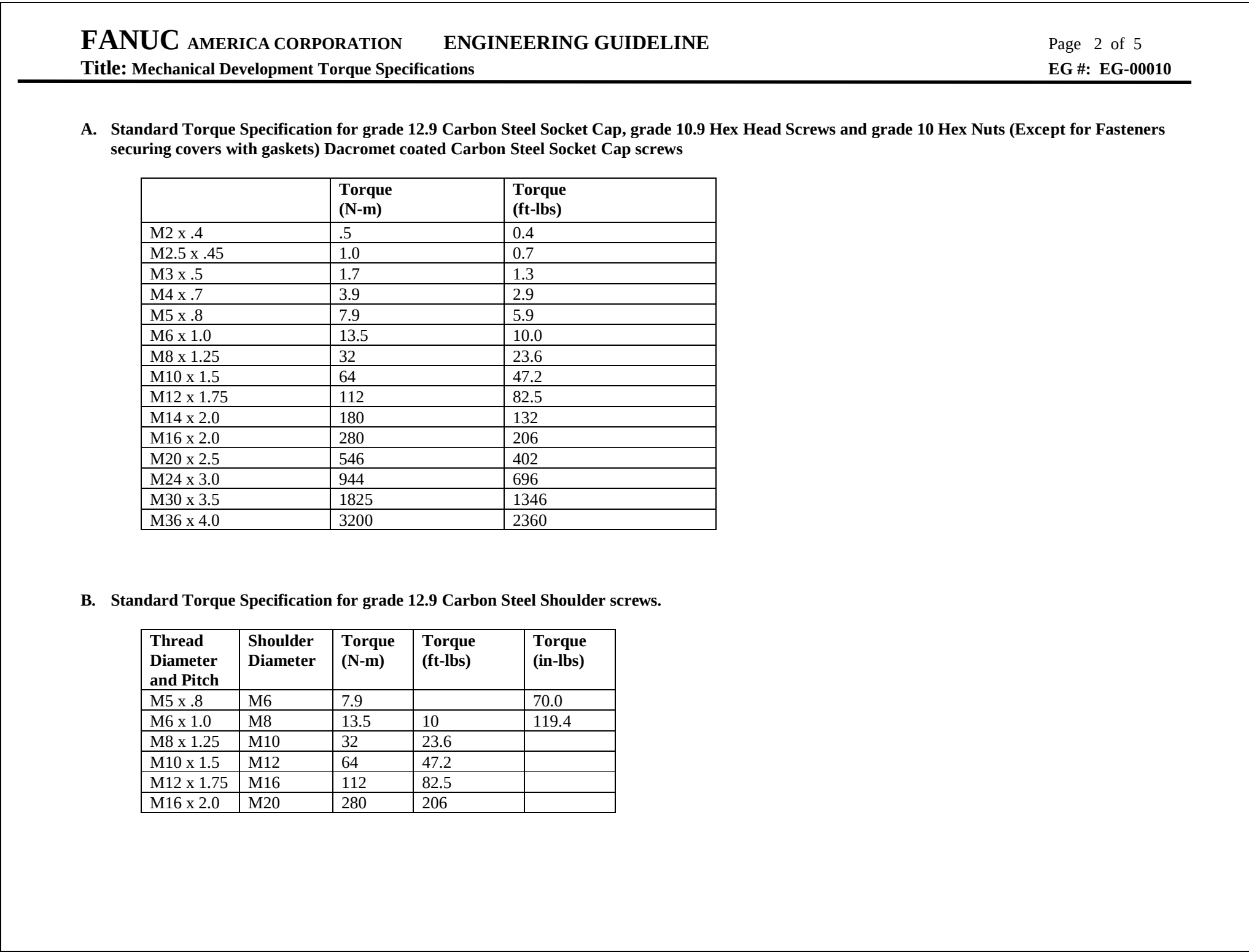


Figure 4-13 EG-00010 Sheet 3 of 5, TORQUE SPECIFICATION

FANUC AMERICA CORPORATION

ENGINEERING GUIDELINE

Title: Mechanical Development Torque Specifications

Page 3 of 5
EG #: EG-00010

C. Standard Torque Specification for all Stainless fasteners and Nuts (Except for Fasteners securing covers with gaskets)

	Torque (N-m)	Torque (ft-lbs)
M2 x .4	0.33	0.25
M2.5 x .45	0.66	0.5
M3 x .5	1.1	0.8
M4 x .7	2.6	2.0
M5 x .8	5.4	4.0
M6 x 1.0	9.2	6.8
M8 x 1.25	22.0	16.2
M10 x 1.5	43.5	32.0
M12 x 1.75	76.0	56.0
M14 x 2.0	121	89.0
M16 x 2.0	189	138
M20 x 2.5	368	271
M24 x 3.0	636	469
M30 x 3.5	1263	932
M36 x 4.0	2207	1628

D. Torque Specification for all Metric Carbon and Stainless Fasteners securing Covers with Gaskets

	Torque (N-m)	Torque (ft-lbs)
M4 x .7	2.0	1.5
M5 x .8	4.0	3.0
M6 x 1.0	6.3	4.6
M8 x 1.25	7.5	5.5
M10 x 1.5	8.4	6.2

Figure 4-14 EG-00010 Sheet 4 of 5, TORQUE SPECIFICATION

FANUC AMERICA CORPORATION ENGINEERING GUIDELINE

Title: Mechanical Development Torque Specifications

Page 4 of 5

EG #: EG-00010

E. Standard Torque Specification for grade 8.8 Carbon Steel Hex Head screws, grade 10.9 Carbon steel Flat Head Socket Screws, grade 10.9 Button Head Socket Screws, grade 8.8 Carbon Steel Low Head Socket Cap Screws, grade 6 Carbon Steel Hex Nuts and all Carbon Steel Jam Nuts (Except for Fasteners securing covers with gaskets), grade 12.9 Zinc plated Carbon Steel Socket Cap screws

	Torque (N-m)	Torque (ft-lbs)
M2 x .4	0.4	0.3
M2.5 x .45	0.7	0.5
M3 x .5	1.0	0.7
M4 x .7	2.2	1.6
M5 x .8	4.3	3.1
M6 x 1.0	7.5	5.5
M8 x 1.25	18.0	13.2
M10 x 1.5	36.0	26.5
M12 x 1.75	67.0	49.4
M14 x 2.0	106.0	78.0
M16 x 2.0	161.0	118.5
M20 x 2.5	315.0	232
M24 x 3.0	544.0	401
M30 x 3.5	1080.0	796
M36 x 4.0	1890.0	1394

F. Standard Torque Specification for all Nylon fasteners and Nuts

	Torque (N-m)	Torque (ft-lbs)
M3 x .5	0.1	0.07
M4 x .7	0.2	0.15
M5 x .8	0.5	0.37
M6 x 1.0	1.0	0.74
M8 x 1.25	2.0	1.4
M10 x 1.5	3.0	2.2
M12 x 1.75	4.0	3.0
M14 x 2.0	6.0	4.4
M16 x 2.0	7.5	5.5

Figure 4-15 EG-00010 Sheet 5 of 5, TORQUE SPECIFICATION

FANUC AMERICA CORPORATION

ENGINEERING GUIDELINE

Title: Mechanical Development Torque Specifications

Page 5 of 5
EG #: EG-00010

G. Standard Torque Specification for all Brass fasteners and Nuts

	Torque (N-m)	Torque (ft-lbs)
M2 x .4	0.14	0.10
M2.5 x .45	0.29	0.21
M3 x .5	0.5	0.37
M4 x .7	1.2	0.9
M5 x .8	2.2	1.6
M6 x 1.0	3.9	2.9
M8 x 1.25	9.0	6.6
M10 x 1.5	17.0	12.5

4.3 Gearhead Mounting Instructions - EG-00316

Figure 4-16 EG-00316 Sheet 1 of 13, GEAR HEAD MOUNTING INSTRUCTIONS

BAYSIDE PX AND PS GEAR REDUCER

Gearhead Mounting Instructions

Featuring ... Patented ServoMount Design for Optimum Servo Performance!!!

Congratulations on your purchase of a Stealth Planetary Gearhead with the exclusive ServoMount mounting system! The following instructions will assist you in mounting the gearhead to your motor. If additional assistance is required, please call Bayside at (516) 484-5353, Monday through Friday, 8:00 am - 7:00 pm EST or visit our website at www.w.baysidemotion.com.

Parts Supplied

- | Bayside Precision Gearhead with ServoMount option (see Fig. 1)
- | Socket Head Cap Screws (4) with Lock Washers
- | Allen Wrench

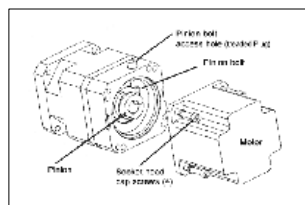


Fig. 1

Mounting Instructions

- Step 1. Remove the threaded plug from the bolt access hole in the rear housing of the gearhead.
- Step 2. Rotate the pinion until the head of the pinion bolt is aligned with the pinion screw access hole.
- Step 3. Insert the supplied Allen wrench through the pinion bolt access hole into the head of the pinion bolt. This bolt should be loose. Do not tighten it. (Fig 2) NOTE: If the motor shaft has a flat, rotate the motor shaft so that the flat will be opposite to the pinion bolt.
- Step 4. For easy mounting, position the motor vertically with the shaft pointing upward. Insert the motor shaft into the gearhead pinion. Align the motor flange mounting holes on the gearhead flange. Motor shaft should be cleaned and dried for best installation. (Fig.3)
- Step 5. Secure the gearhead to the motor using the (4) socket head cap screws and (4) lock washers supplied. (Fig. 4)
- Step 6. With the gearhead fully seated onto the motor flange, back off the (4) socket head cap screws one complete turn.
- Step 7. Refer to the torque specification from the following table. Using the supplied Allen wrench, tighten the pinion bolt to the noted torque specification.
- Step 8. Fully tighten the (4) socket head cap screws.
- Step 9. Reinsert and tighten the threaded plug in the pinion bolt access hole.



Fig. 2



Fig. 3



Fig. 4

See table on reverse side for screw tightening torques.

BAYSIDE
MOTION GROUP

27 Seaview Boulevard * Port Washington, NY 11050 * (516-484-5353 * Fax: 516-484-5496
baysidemotion.com

rev 3.0 / 203
PN: 12197018

Frame Size	Motor Shaft Diameter		Screw Size	Tightening Torque	
(mm)	inches	(mm)		in-lbs.	(Nm)
40	< .250	(< 6.35)	2.5	11	1.21
60	.230 - .630	(5.8 - 16)	M4	41	4.6
90	.250 - .630	(6.4 - 16.0)	M4	41	4.6
	.630 - .748	(16.0 - 19.0)	M5	84	9.5
115	.375 - .751	(9.5 - 19.1)	M5	84	9.5
	.787 - .946	(20.0 - 24.0)	M6	140	16
142	.500 - .945	(12.7 - 24.0)	M6	140	16
	1.00 - 1.379	(25.4 - 35.0)	M8	345	39
180	.624 - 1.378	(15.8 - 35.0)	M8	345	39
	1.496 - 1.654	(38.0 - 42.0)	M10	681	77
220	.945 - 1.388	(25.4 - 35.2)	M10	681	77
	1.496 - 2.165	(38.0 - 55.0)	M12	1195	135
300	All	All	M12	1195	135

NOTE: APPLY LOCTITE TO THE THREADS OF THE PINION BOLT. USE LOCTITE 242 FOR SCREW SIZES ABOVE M6, AND LOCTITE 222 FOR SCREWS SIZE M6 AND BELOW

WARNING

Some components are heavy. Use extreme care and follow proper lifting procedures when you install or remove components. Otherwise you could injure personnel or damage equipment.

WARNING

Always disconnect power to the robot before you perform any service or maintenance procedures. Otherwise, you could injure personnel or damage equipment.

UNLESS OTHERWISE NOTED:
SEE EG-00008 FOR LOCTITE REQUIREMENTS
SEE EG-00010 FOR TORQUE SPECIFICATIONS

Figure 4-17 EG-00316 Sheet 2 of 13, GEAR HEAD MOUNTING INSTRUCTIONS

WITTENSTEIN (ALPHA) CP GEAR REDUCER

Wittenstein (Alpha) Gear Reducers - CP models

Motor mounting

All gear reducers are treated with an anti-corrosion agent at the gear input and output.

Before mounting the gear reducer, remove any trace of anti-corrosion agent. Use a clean cloth moistened with a suitable cleaning agent (fat dissolving but not aggressive).

Use calibrated torque wrenches for the respective torque range.

A setscrew and a slit spacer sleeve connect the motor shaft and the gear drive shaft

Clean the mounting surfaces of the motor and gear reducer with a clean cloth. Do not use compressed air. Check the surfaces for impurities and damage. Do not put any devices into operation that show evidence of damage.

Turn the clamping hub so that the setscrew can be reached through the mounting holes in the adapter flange.

Take care that the slot of the spacer sleeve is positioned 90° to the setscrew. Readjust the spacer sleeve if necessary.

Make sure that the motor allows itself to be moved into position "easily". Install the motor vertically.

Position the motor so that the mounting surfaces fit together

There is to be no gap between the motor and adapter flange.

Coat the motor mounting bolts with Loctite 242 and screw the motor to the adapter flange. Torque motor mounting bolts.

Turn setscrew in the clamping hub to the tightening torque per chart.

Tightening Torque (N-M)	Screw Size (mm)	Hex Bit Size (mm)
6	M4	3
14	M5	4
23	M6	5
45	M8	6

Install set screw cover plug

Output Shaft Mounting

Mount gears, pulleys or drive shafts to output without forcing. Never force or hammer.

During assembly, do not exceed the axial force per table below.

Gear reducer size	040	060	080	115
Max Force (N)	230	750	1600	2100



WARNING
Some components are heavy. Use extreme care and follow proper lifting procedures when you install or remove components. Otherwise you could injure personnel or damage equipment



WARNING
Always disconnect power to the robot before you perform any service or maintenance procedures. Otherwise, you could injure personnel or damage equipment.

UNLESS OTHERWISE NOTED:
SEE EG-00008 FOR LOCTITE REQUIREMENTS
SEE EG-00010 FOR TORQUE SPECIFICATIONS

Figure 4-18 EG-00316 Sheet 3 of 13, GEAR HEAD MOUNTING INSTRUCTIONS

ZF GEAR REDUCER

Assembly Instructions
ZF-Servoplan PGE - Planetary Gearboxes



ZF Maschinenantriebe GmbH
ZF Industrial Drives

General:

A torsionally rigid and backlash free connection between motor and gearbox can quickly and easily be achieved by using the following assembly instructions.
Note: Please only use motors with a flange face and run out accuracy class DIN 42955 N or DIN 42955 R.

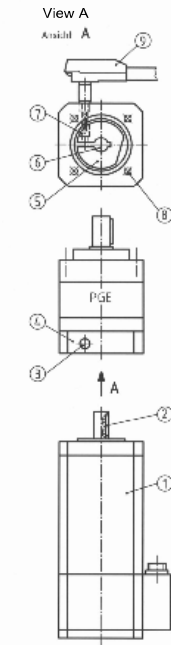
To avoid misalignment inside the power train, the PGE utilizes our new "open centering" design making the laborious task of centering with the flange and pilot obsolete. Simply slip the gearbox hollow shaft (6) over the motor shaft, tighten the clamping bolt (7) on the compression coupling (5), and attach the gearbox to the motor using the mounting holes (10).

The PGE planetary gearboxes are maintenance free and have lifetime lubrication.

Motor Mounting Instructions:

1. If the motor shaft has a key (2), remove it. For applications with motor speeds over 3000 rpm, we recommend using a half key to completely fill the key way. The key may not stand out above the height of the motor shaft.
2. Remove vent plug (3) on the adapter flange (4) and put it aside, reserving it for later use.
3. Turn the compression coupling (5) until the head of the clamping bolt (7) is visible through the access hole.
4. If a PGE with an adapter flange (4) screwed on, is not directly mounted on the motor by using the adapter flange (4), then the adapter flange (4) must be removed from the PGE. Remove four hexagonal socket-head screws (8), then pull off the adapter flange (4) from the PGE.
5. Examine all contact surfaces on the motor flange and the adapter flange (4) to ensure they are clean and grease free. Also inspect the motor shaft and the hub to ensure they are free from damage (burrs, scoring, etc.).
6. Place motor (1) vertically on the work surface and secure it from falling. Ensure that the key way in the motor shaft is positioned opposite the slit (180°) in the compression coupling when assembled.
7. If a reduction bushing is used, ensure the slit in the bushing is aligned with the compression coupling slit.
8. If motor installation is as described under 4., then the previously removed adapter flange (4) must be placed onto the motor flange surface and then screwed on.
9. If the PGE is too heavy to lift manually, a suitable hoisting machine should be used. Slip gearbox hollow shaft (6) over motor shaft until the adapter flange mates with the motor flange over its entire surface. Do not strike the gearbox or use excessive force to ensure a good fit.
10. Tighten the clamping bolt (7) to the prescribed torque (see table tightening torque) using a suitable torque wrench (9).
11. Now bolt PGE with adapter flange (4) directly on motor (1) or if you have a motor installation as described under 4., bolt the PGE onto adapter flange (4). Screws (8) must be diagonally transferred and evenly tightened.
12. Then bolt PGE and motor together using the mounting holes (8) in the adapter flange. The bolts must be diagonally transferred and uniformly tightened.
13. Insert access hole plug (3) into adapter flange (4).

tightening torque for the clamping bolt (7)	
clamping screw DIN 912	tightening torque
M3	2.1 Nm
M4	4.2 Nm
M5	8.3 Nm
M8	43 Nm



Note: Failure to follow assembly instructions may lead to gearbox and/or motor damage and will void any warranty either explicit or implied.

4172 758 101a
Version:19.07.2004

WARNING
Some components are heavy. Use extreme care and follow proper lifting procedures when you install or remove components. Otherwise you could injure personnel or damage equipment.

WARNING
Always disconnect power to the robot before you perform any service or maintenance procedures. Otherwise, you could injure personnel or damage equipment.

UNLESS OTHERWISE NOTED:
SEE EG-00008 FOR LOCTITE REQUIREMENTS
SEE EG-00010 FOR TORQUE SPECIFICATIONS

Figure 4-19 EG-00316 Sheet 4 of 13, GEAR HEAD MOUNTING INSTRUCTIONS

ZF GEAR REDUCER

Gearbox Output Shaft

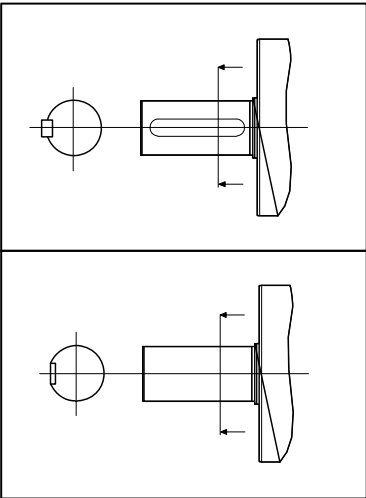
The gearbox output shaft is available in the following designs:

Standard:
Output shaft with key as per DIN 6885 sheet 1 (08.68 issue) for keyed shaft-hub connections. This type of connection is suitable for constant direction, where applications, requirements are not as stringent. This connection type requires additional axial fixing of the hub. A centering bore with thread is provided on the face end of the gearbox output shaft for this purpose (as per DIN 332, sheet 2).

Options: fitted with a half key.

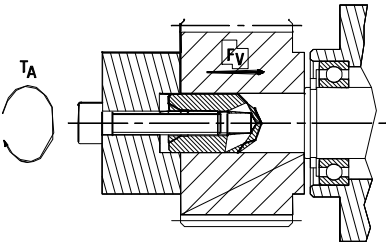
Plain output shaft on special request.

OUTPUT SHAFT MOUNTING



Size:							
		PGE 12/1	PGE 25/1	PGE 50/1	PGE 100/1	PGE 200/1	PGE 500/1
		12/2	25/2	50/2	100/2	200/2	500/2
		12/3	25/3	50/3	100/3	200/3	500/3
Thread	DIN 332	M4	M5	M6	M10	M16	M20
TA	[Nm]	3.4	5.5	9.5	46	190	380
FV	[kN]	4.1	6.5	9.5	27	74	110

TA = Tightening torque
FV = Pretensioning Force



WARNING

Some components are heavy. Use extreme care and follow proper lifting procedures when you install or remove components. Otherwise you could injure personnel or damage equipment.

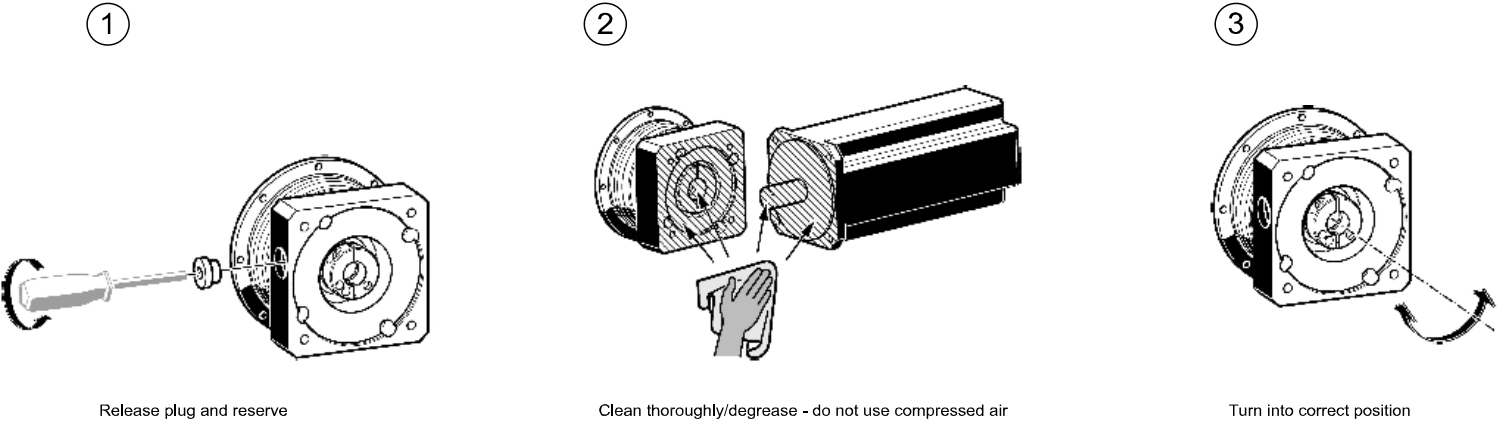
WARNING

Always disconnect power to the robot before you perform any service or maintenance procedures. Otherwise, you could injure personnel or damage equipment.

UNLESS OTHERWISE NOTED:
SEE EG-00008 FOR LOCTITE REQUIREMENTS
SEE EG-00010 FOR TORQUE SPECIFICATIONS

Figure 4-20 EG-00316 Sheet 5 of 13, GEAR HEAD MOUNTING INSTRUCTIONS

WITTENSTEIN (ALPHA) TP+ GEAR REDUCER



WARNING
Some components are heavy. Use extreme care and follow proper lifting procedures when you install or remove components. Otherwise you could injure personnel or damage equipment.

WARNING
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UNLESS OTHERWISE NOTED:
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SEE EG-00010 FOR TORQUE SPECIFICATIONS

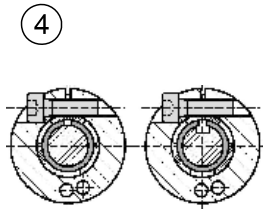
Figure 4-21 EG-00316 Sheet 6 of 13, GEAR HEAD MOUNTING INSTRUCTIONS

WITTENSTEIN (ALPHA) TP+ GEAR REDUCER

WARNING
Some components are heavy. Use extreme care and follow proper lifting procedures when you install or remove components. Otherwise you could injure personnel or damage equipment

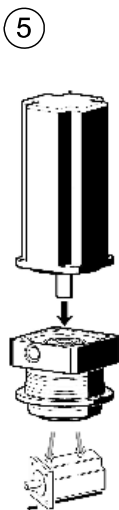
WARNING
Always disconnect power to the robot before you perform any service or maintenance procedures. Otherwise, you could injure personnel or damage equipment.

UNLESS OTHERWISE NOTED:
SEE EG-00008 FOR LOCTITE REQUIREMENTS
SEE EG-00010 FOR TORQUE SPECIFICATIONS

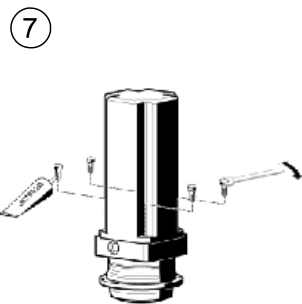
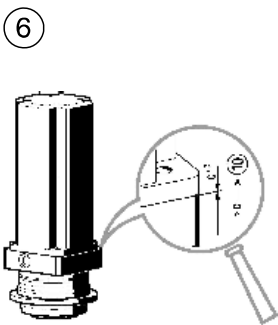


With grooved shaft, remove feather key. Mount slot, spacer sleeve, and feather key groove in the correct position.

Note: Spacer sleeve may not be present for some applications. Use of a spacer sleeve is dependent on motor shaft diameter.



CAUTION! If the motor does not move into position easily, loosen the screw in fig. 4 by one rotation.



SW: Width across flats
TA: Tightening torque

Figure 4-22 EG-00316 Sheet 7 of 13, GEAR HEAD MOUNTING INSTRUCTIONS

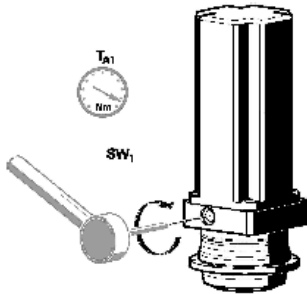
WITTENSTEIN (ALPHA) TP+ GEAR REDUCER

WARNING
Some components are heavy. Use extreme care and follow proper lifting procedures when you install or remove components. Otherwise you could injure personnel or damage equipment

WARNING
Always disconnect power to the robot before you perform any service or maintenance procedures. Otherwise, you could injure personnel or damage equipment.

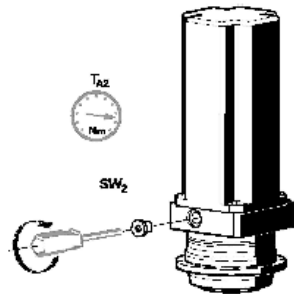
UNLESS OTHERWISE NOTED:
SEE EG-00008 FOR LOCTITE REQUIREMENTS
SEE EG-00010 FOR TORQUE SPECIFICATIONS

8



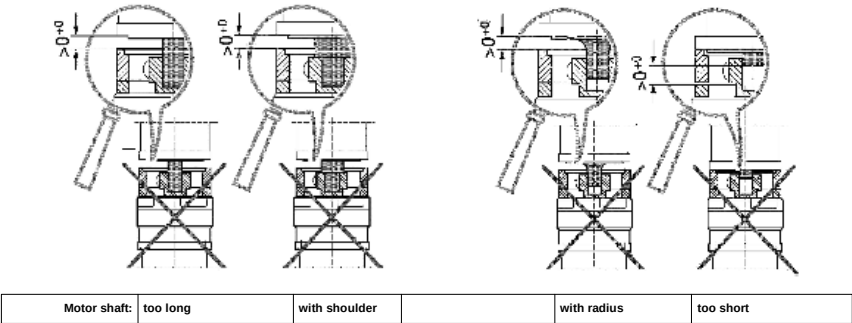
SCREW ϕ [mm]	HEX SIZE	TORQUE
	SW ₁ [mm]	T _{A1} [Nm]
4	3	4.1
5	4	9.5
6	5	16
8	6	39
10	8	79
12	10	135
16	14	330

9



SCREW ϕ [mm]	HEX SIZE	TORQUE
	SW ₂ [mm]	T _{A2} [Nm]
6	5	10
10	8	35
12	10	50
14	12	70

10



CAUTION! In these cases, look up instructions in the operating manual

Figure 4-23 EG-00316 Sheet 8 of 13, GEAR HEAD MOUNTING INSTRUCTIONS

WITTENSTEIN (ALPHA) SP+ GEAR REDUCER

WARNING

Some components are heavy. Use extreme care and follow proper lifting procedures when you install or remove components. Otherwise you could injure personnel or damage equipment.

WARNING

Always disconnect power to the robot before you perform any service or maintenance procedures. Otherwise, you could injure personnel or damage equipment.

UNLESS OTHERWISE NOTED:
SEE EG-00008 FOR LOCTITE REQUIREMENTS
SEE EG-00010 FOR TORQUE SPECIFICATIONS

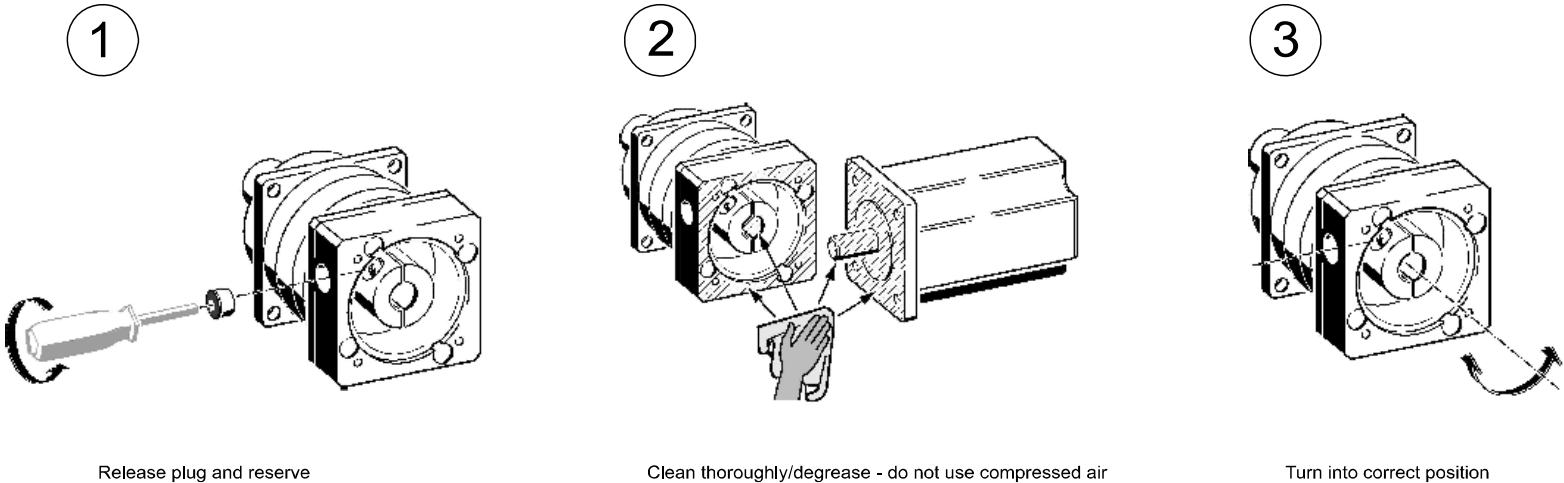


Figure 4-24 EG-00316 Sheet 9 of 13, GEAR HEAD MOUNTING INSTRUCTIONS

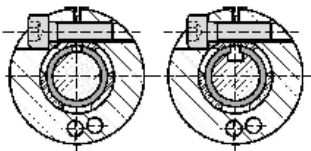
WITTENSTEIN (ALPHA) SP+ GEAR REDUCER

WARNING
Some components are heavy. Use extreme care and follow proper lifting procedures when you install or remove components. Otherwise you could injure personnel or damage equipment.

WARNING
Always disconnect power to the robot before you perform any service or maintenance procedures. Otherwise, you could injure personnel or damage equipment.

UNLESS OTHERWISE NOTED:
SEE EG-00008 FOR LOCTITE REQUIREMENTS
SEE EG-00010 FOR TORQUE SPECIFICATIONS

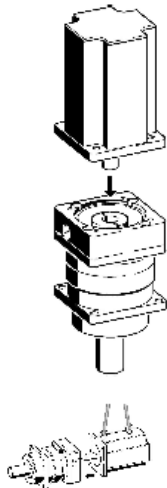
4



With grooved shaft, remove feather key. Mount slot, spacer sleeve, and feather key groove in the correct position.

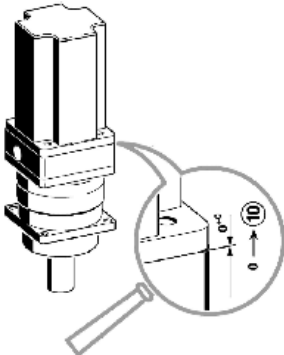
Note: Spacer sleeve may not be present for some applications. Use of a spacer sleeve is dependent on motor shaft diameter.

5

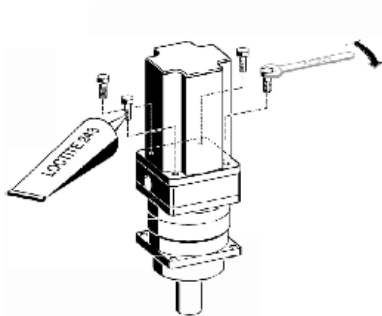


CAUTION! If the motor does not move into position easily, loosen the screw in fig. 4 by one rotation.

6



7



SW: Width across flats
TA: Tightening torque

Figure 4-25 EG-00316 Sheet 10 of 13, GEAR HEAD MOUNTING INSTRUCTIONS

WITTENSTEIN (ALPHA) SP+ GEAR REDUCER

8

SCREW ϕ [mm]	HEX SIZE SW ₁ [mm]	TORQUE T _{A1} [Nm]
4	3	4.1
5	4	9.5
6	5	14
8	6	35
10	8	79
12	10	135
16	14	330

9

SCREW ϕ [mm]	HEX SIZE SW ₂ [mm]	TORQUE T _{A2} [Nm]
6	5	10
10	8	35
12	10	50
14	12	70

10

Motor shaft:	too long	with shoulder	with radius	too short
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CAUTION! In these cases, look up instructions in the operating manual

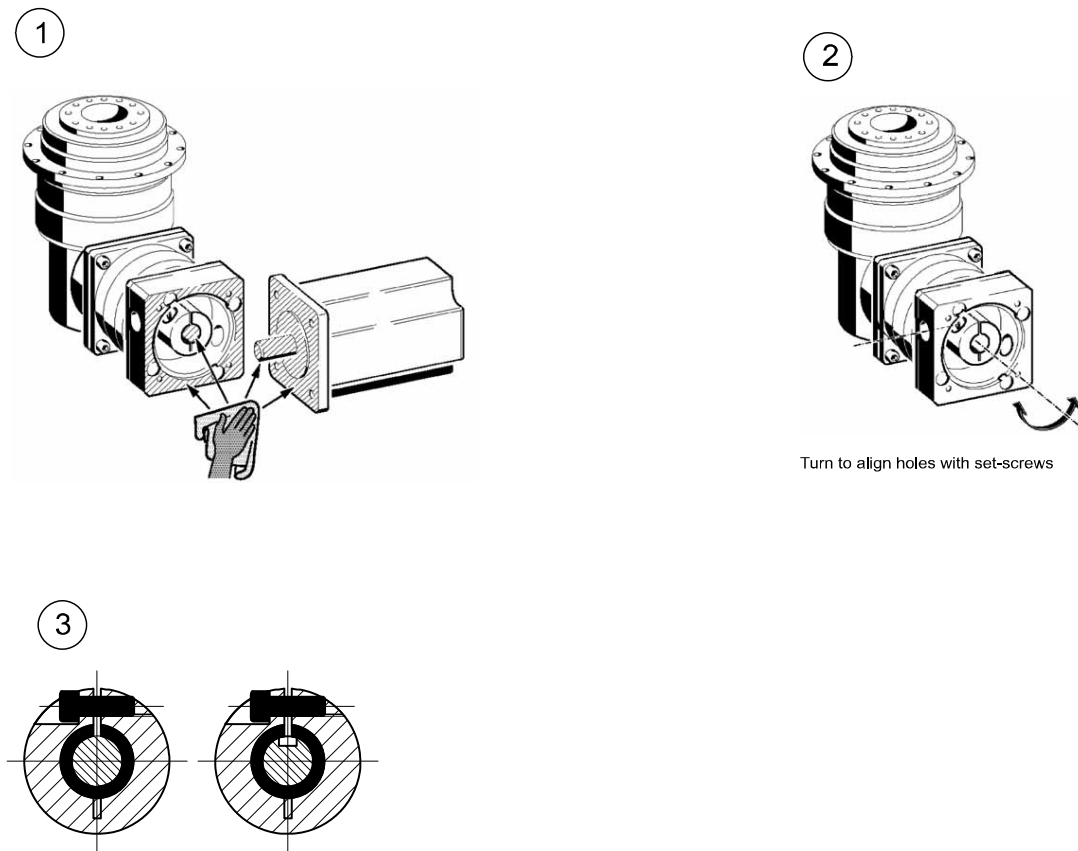
WARNING
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WARNING
Always disconnect power to the robot before you perform any service or maintenance procedures. Otherwise, you could injure personnel or damage equipment.

UNLESS OTHERWISE NOTED:
SEE EG-00008 FOR LOCTITE REQUIREMENTS
SEE EG-00010 FOR TORQUE SPECIFICATIONS

Figure 4-26 EG-00316 Sheet 11 of 13, GEAR HEAD MOUNTING INSTRUCTIONS

WITTENSTEIN (ALPHA) TPK GEAR REDUCER



Turn to align holes with set-screws

For keywayed motorshaft, mount slot and keyway in proper position.

WARNING

Some components are heavy. Use extreme care and follow proper lifting procedures when you install or remove components. Otherwise you could injure personnel or damage equipment.

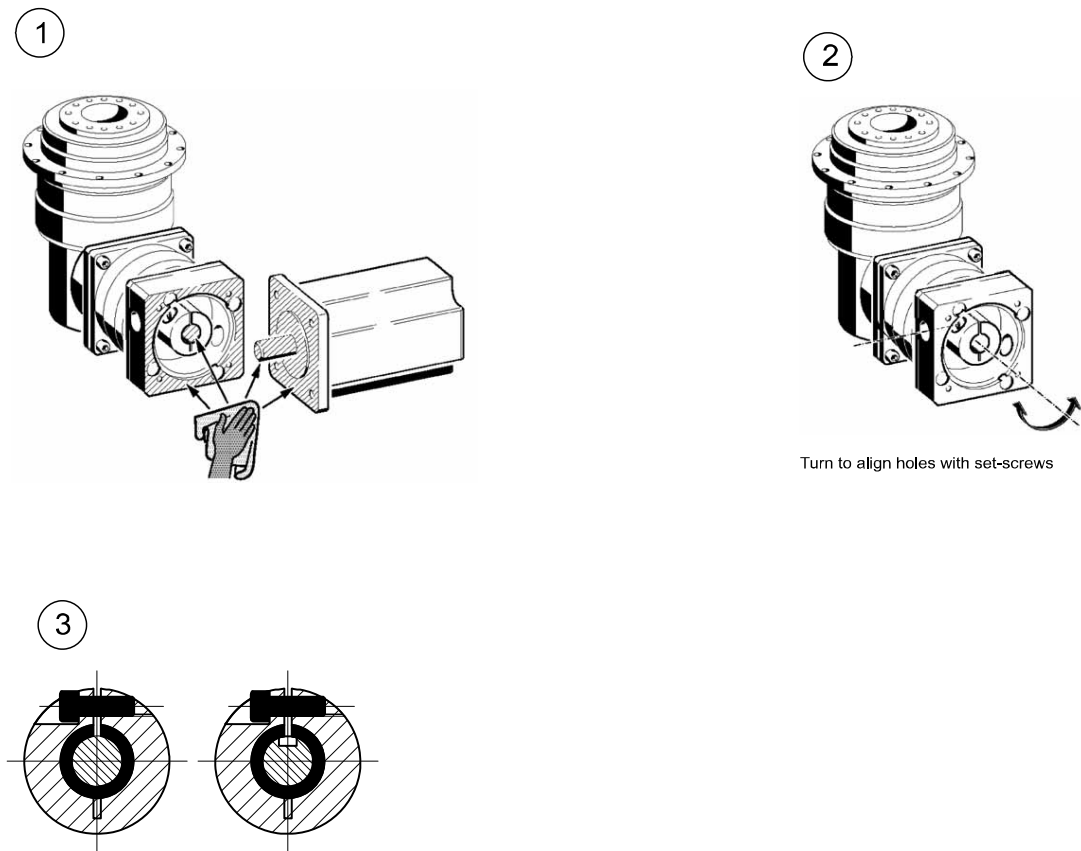
WARNING

Always disconnect power to the robot before you perform any service or maintenance procedures. Otherwise, you could injure personnel or damage equipment.

UNLESS OTHERWISE NOTED:
SEE EG-00008 FOR LOCTITE REQUIREMENTS
SEE EG-00010 FOR TORQUE SPECIFICATIONS

Figure 4-27 EG-00316 Sheet 12 of 13, GEAR HEAD MOUNTING INSTRUCTIONS

WITTENSTEIN (ALPHA) TPK GEAR REDUCER



Turn to align holes with set-screws

For keywayed motorshaft, mount slot and keyway in proper position.

WARNING

Some components are heavy. Use extreme care and follow proper lifting procedures when you install or remove components. Otherwise you could injure personnel or damage equipment.

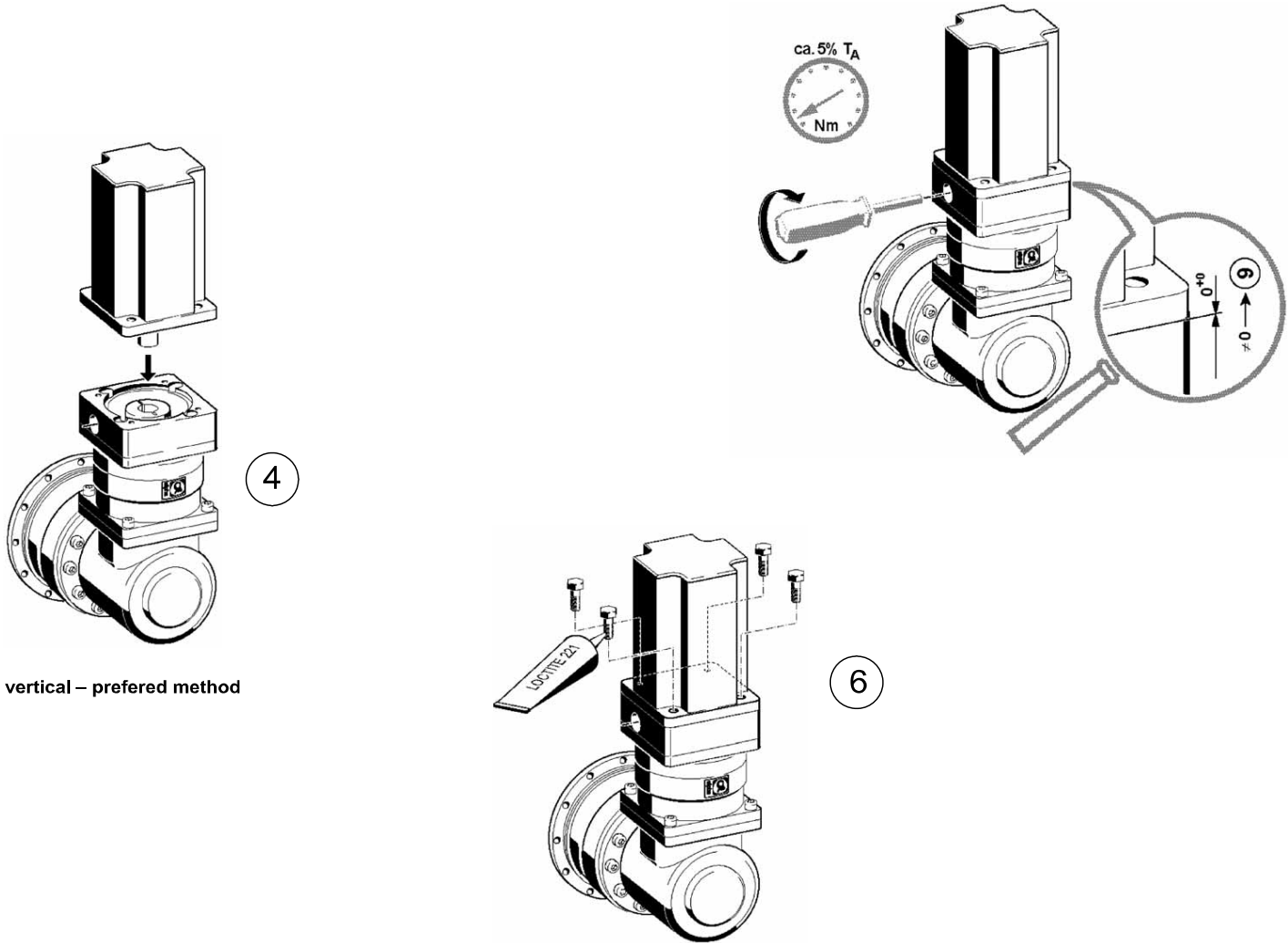
WARNING

Always disconnect power to the robot before you perform any service or maintenance procedures. Otherwise, you could injure personnel or damage equipment.

UNLESS OTHERWISE NOTED:
SEE EG-00008 FOR LOCTITE REQUIREMENTS
SEE EG-00010 FOR TORQUE SPECIFICATIONS

Figure 4-28 EG-00316 Sheet 13 of 13, GEAR HEAD MOUNTING INSTRUCTIONS

WITTENSTEIN (ALPHA) TPK GEAR REDUCER



WARNING
Some components are heavy. Use extreme care and follow proper lifting procedures when you install or remove components. Otherwise you could injure personnel or damage equipment.

WARNING
Always disconnect power to the robot before you perform any service or maintenance procedures. Otherwise, you could injure personnel or damage equipment.

UNLESS OTHERWISE NOTED:
SEE EG-00008 FOR LOCTITE REQUIREMENTS
SEE EG-00010 FOR TORQUE SPECIFICATIONS